

Diffusion Estimation and MSD/VACF Consistency Specification

G2/GK3 after H0: Explicit Plateau Selection and Signature-Checked Comparison

mdstats

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1 Purpose and status

This document specifies the H0-hardened functions in

`mdstats/analysis/diffusion.py`

`estimate_diffusion_plateau(...)`

`compare_msd_vacf_diffusion(...)`

and the immutable `DiffusionEstimate` and `DiffusionComparisonResult` schemas. The module consumes existing MSD and running Green-Kubo results. It does not recompute trajectories, correlations, or transport integrals.

2 Borrowed theory and numerical methods

For a physical subspace of rank d , the Einstein relation gives [1]

$$M_B(t) \sim 2dD_B t.$$

Thus a linear slope m_B yields

$$D_B = \frac{m_B}{2d}.$$

The compared VACF estimate follows Green [2] and Kubo [3]. The centered ordinary-least-squares formulas used for descriptive slope diagnostics are standard regression algebra. Explicit interval selection, refusal to infer an uncertainty from adjacent running-integral samples, semantic-signature checking, and the symmetric relative difference are mdstats policies.

3 Explicit plateau estimate

3.1 Public function

```
def estimate_diffusion_plateau(
    running: VACFDiffusionResult,
    *,
    time_range_ps: tuple[float, float] | None = None,
    minimum_points: int = 8,
    slope_tolerance: float | None = None,
    method: Literal["explicit", "stable_window"] = "explicit",
) -> DiffusionEstimate:
    ...
```

Only `method="explicit"` is implemented. `"stable_window"` raises `NotImplementedError` until a separately specified automatic-window stage.

The requested interval is inclusive within floating-point boundary tolerance and must lie inside stored lag times. At least `minimum_points` selected samples are required.

3.2 Uniform-grid requirement

The current estimator is the arithmetic mean

$$\widehat{D} = \frac{1}{N} \sum_{j=1}^N D(t_j).$$

H0 requires the selected lag samples to be uniformly spaced. If their increments are not equal within strict tolerance, the function raises. It does not allow a dense region of an irregular grid to receive more implicit weight. A future time-weighted interval estimator must use another method name.

3.3 Diagnostics

The function stores descriptive values over the selected interval:

- mean, median, sample standard deviation, minimum, maximum, and span;
- endpoint drift;
- centered linear slope and intercept;
- R^2 and residual RMS;
- selected sample spacing; and
- an optional pass/fail result for the absolute slope tolerance.

`standard_error_a2_per_ps` remains `None`. Adjacent values on one cumulative curve are serially correlated and are not treated as independent replicas.

4 Diffusion estimate result

```
@dataclass(frozen=True, slots=True)
class DiffusionEstimate:
    value_a2_per_ps: float
    standard_error_a2_per_ps: float | None
    time_range_ps: tuple[float, float]
    method: str
    component: str
    dimensions: int
    n_points: int
    is_stable: bool | None
    diagnostics: Mapping[str, Any]
    metadata: Mapping[str, Any]
    projection_basis: NDArray[np.float64]
    projection_labels: tuple[str, ...] | None
    signature: DynamicsInputSignature | None
```

The basis rank equals `dimensions`. The component label and optional signature must agree with that subspace. Diagnostics and metadata are recursively immutable. The cm²/s property uses the exact package conversion factor 10⁻⁴.

5 MSD/VACF comparison

5.1 Public function

```
def compare_msd_vacf_diffusion(
    msd: MSDResult,
    vacf_diffusion: DiffusionEstimate,
    *,
    msd_fit_range_ps: tuple[float, float],
    dimensions: Literal[1, 2, 3] | None = None,
) -> DiffusionComparisonResult:
    ...
```

The optional `dimensions` argument is deprecated and acts only as a consistency check against the stored subspace rank.

5.2 Admission contract

The comparison requires:

- `msd.mode == "time_averaged"`;
- `msd.coordinate_mode == "laboratory"`;
- complete `DynamicsInputSignature` objects on both inputs; and
- equality of every semantic field after assigning the VACF estimate subspace to the MSD signature.

This detects mismatches in source identity, exact frame sequence and times, measured atoms, coordinate/reference-cell treatment, drift mode, exact drift- reference atoms, velocity source, and physical projection.

Unsigned legacy results fail closed.

5.3 Projected MSD fit

The selected VACF estimate basis is also used for the MSD:

$$M_B(t) = \text{tr}[BM(t)B^T].$$

Canonical axis subsets may use stored components. A rotated basis requires the full MSD tensor. At least three stored MSD points are required. Centered OLS with an intercept produces slope m_B , then

$$D_{\text{MSD}} = \frac{m_B}{2d}.$$

5.4 Difference measures

The result stores

$$\Delta D = D_{\text{MSD}} - D_{\text{VACF}},$$

$$|\Delta D|,$$

and the symmetric relative difference

$$\delta_{\text{sym}} = \begin{cases} 0, & |D_{\text{MSD}}| + |D_{\text{VACF}}| = 0, \\ \frac{2|D_{\text{MSD}} - D_{\text{VACF}}|}{|D_{\text{MSD}}| + |D_{\text{VACF}}|}, & \text{otherwise.} \end{cases}$$

Neither estimator is declared authoritative.

6 Comparison result

```
@dataclass(frozen=True, slots=True)
class DiffusionComparisonResult:
    msd_diffusion_a2_per_ps: float
    vacf_diffusion_a2_per_ps: float
    signed_difference_a2_per_ps: float
    absolute_difference_a2_per_ps: float
    symmetric_relative_difference: float
    msd_fit_range_ps: tuple[float, float]
    msd_slope_a2_per_ps: float
    msd_intercept_a2: float
    component: str
    dimensions: int
    n_msd_points: int
    diagnostics: Mapping[str, Any]
    metadata: Mapping[str, Any]
    projection_basis: NDArray[np.float64]
    projection_labels: tuple[str, ...] | None
    signature: DynamicsInputSignature
```

The constructor validates all algebraic difference identities, subspace fields, fit interval, point count, and signature projection. Arrays and nested mappings are immutable.

7 Interpretation policy

The module reports, but does not automatically resolve:

- nonpositive MSD slopes;
- a VACF interval that fails the requested slope tolerance;
- a VACF interval whose stability was not assessed;
- imperfect linearity in the MSD fit; and
- finite-record differences between Einstein and Green-Kubo estimates.

No automatic ballistic/caged/diffusive classification is performed.

8 Required tests

- constant and gently drifting running curves produce expected diagnostics;
- irregular selected lag grids raise;
- boolean point counts and invalid tolerances raise;
- an unsupported automatic window raises explicitly;
- scalar 3D and Cartesian comparison values preserve pre-H0 references;
- axis-subset and rotated MSD projections use the stored VACF subspace;
- a rotated projection without the MSD tensor raises;
- different drift-reference atoms raise;
- different slices under the same filename raise;
- coordinate/reference-cell and velocity-source mismatches raise;
- unsigned results raise;
- the deprecated `dimensions` check cannot reinterpret the subspace;
- difference identities and units are exact; and
- arrays, diagnostics, and metadata are deeply immutable.

9 References

- [1] A. Einstein, *Ann. Phys.* **322**, 549-560 (1905). DOI: 10.1002/andp.19053220806.
- [2] M. S. Green, *J. Chem. Phys.* **22**, 398-413 (1954). DOI: 10.1063/1.1740082.
- [3] R. Kubo, *J. Phys. Soc. Jpn.* **12**, 570-586 (1957). DOI: 10.1143/JPSJ.12.570.